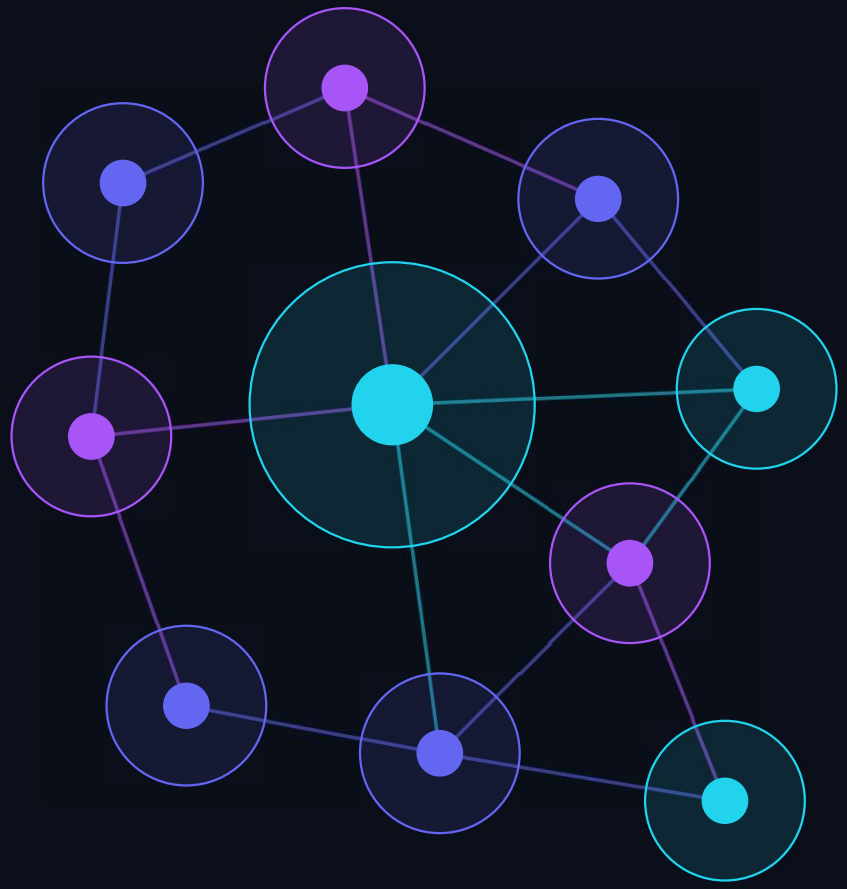
RiskPulse: AI/NLP Financial Risk Engine

Turning unstructured news and social text into actionable risk signals



Ashish Ranjan | VIT Chennai | S&P Global & Crisil Campus Hackathon 2026

Risk teams read too much, too late

THE PROBLEM

Thousands of news and social posts every day.

Manual review is slow and reactive.

One engine, three signals

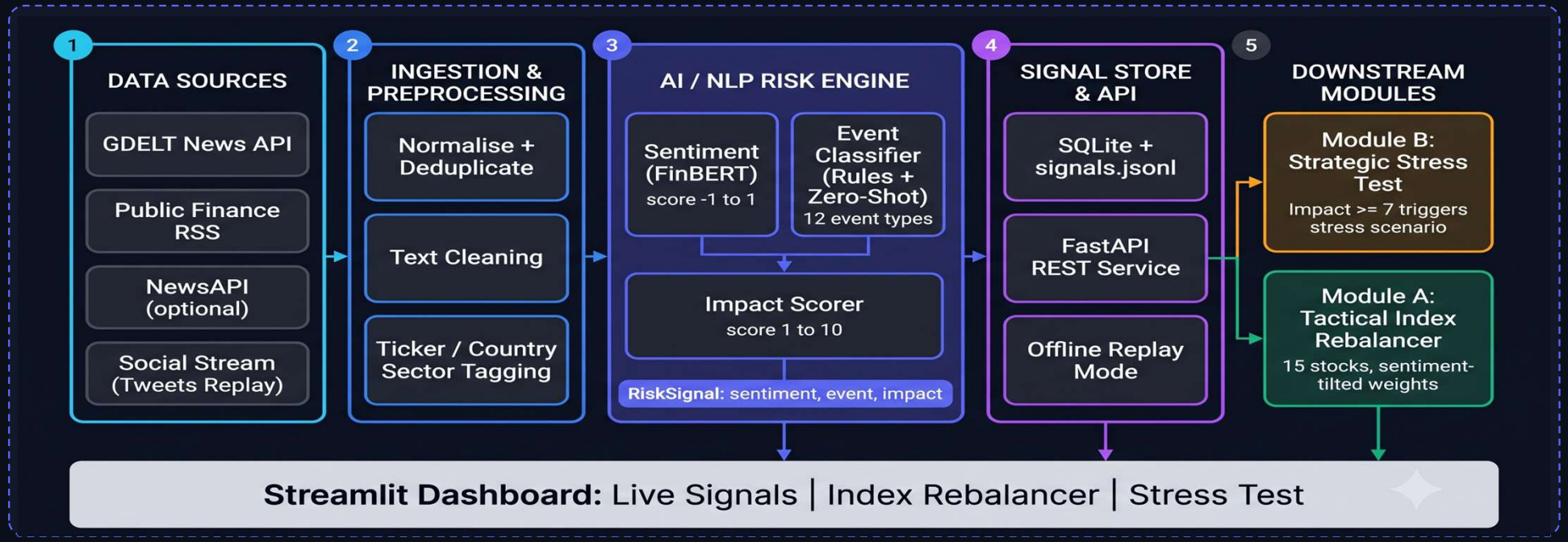
Sentiment | Event | Impact

1 **One NLP engine → 3 outputs**
Sentiment (-1 to 1) · Event type · Impact (1-10)

2 **Signals served via REST API**
Delivered through a REST API and JSONL/SQLite

3 **Two applications**
Module B strategic stress testing (primary) and Module A tactical index rebalancing

System design and data flow



1 Sources
GDELT, RSS, NewsAPI (optional), tweets replay

2 Ingestion
+ entity tagging

3 NLP engine
FinBERT sentiment + event classifier + impact scorer

4 Signal store
+ FastAPI

5 Dashboard
Module A and Module B

Key technical choices, and why

FinBERT

Finance-tuned sentiment

Hybrid event classifier

Rules for precision, zero-shot NLI for coverage (12 labels)

Explainable impact score

Event prior + sentiment + keywords + source credibility + corroboration; components stored in every signal

Event-to-exposure linking

Signal countries/sectors give matching positions a larger shock

Demo-proof

One-command launcher, offline replay mode, keyless live sources

Engineering

Config-driven, 102 automated tests, Ruff-clean, FastAPI docs

Measured results

SENTIMENT ACCURACY (FINBERT)

80.3%

vs 66.4% naive lexicon. Macro-F1 0.808 vs 0.535. N = 1,000 FinancialPhraseBank headlines.

EVENT CLASSIFICATION

55.0%

Accuracy; 0.593 macro-F1 on 12 classes (N = 120). Strongest F1: M&A 0.90, Geopolitical 0.89, Credit Event 0.78.

THROUGHPUT ON CPU

400+ docs/s

About 400+ docs/s on rules/cached; 11-21 docs/s on live transformer passes.

Stress test: $5.02B synthetic book (351 positions)

Scenario	Impact	MTM loss	ECL change
Geopolitical	7.6	-$221.7M (-4.42%)	+$6.67M
Credit event	8.3	-$135.7M (-2.71%)	+$6.73M
Rate hike	7.3	-$77.3M (-1.54%)	+$1.90M

Impact vs next-day |return|: Spearman rho = -0.09, p = 0.41, N = 81. No significant relationship found; impact is a severity heuristic, not a return predictor.

Module A: sentiment-tilted weights under caps and turnover limits, shown over time on the dashboard.



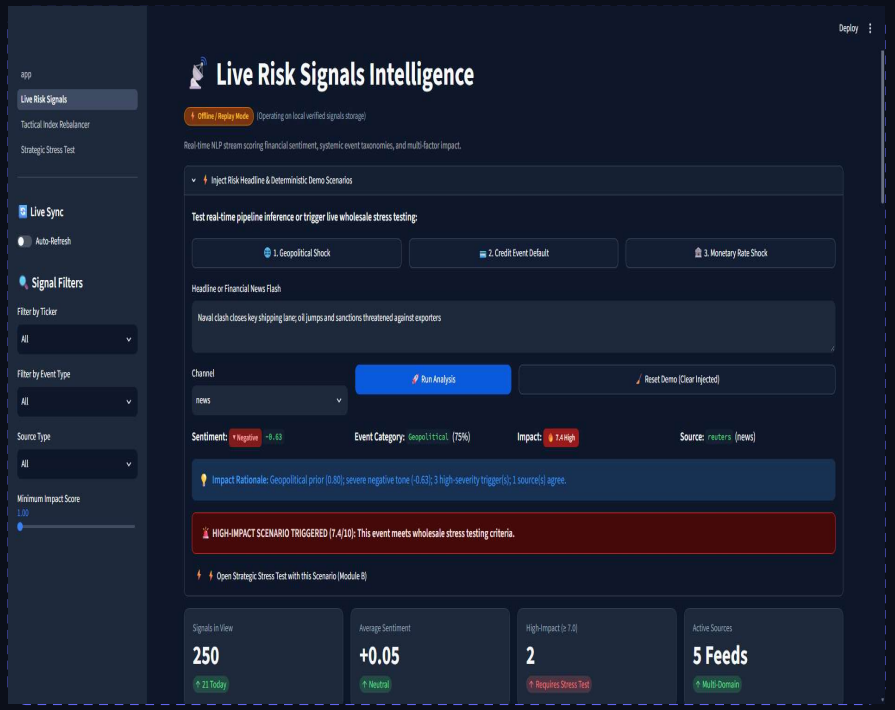
Why it matters

Earlier warning
Text becomes structured risk signals in seconds

Consistent, auditable scenario selection
A threshold (impact ≥ 7) triggers the stress test automatically

Explainability
Every score shows its components, which suits model-risk review

Two use cases, one engine
Banks (balance-sheet stress) and asset managers (tactical weights)



Live signals dashboard

Limitations and next steps

Limitations

- Synthetic portfolio and illustrative shock sizes
- Short price window (N = 81); impact-return link not significant
- Simplified valuation (duration/convexity/DV01/CS01, no full revaluation)
- Zero-shot classification is slow on CPU; social data is a replay

Next steps

- Calibrate shocks to historical episodes and extend validation across multiple regimes
- Distil/quantise models (ONNX) for higher throughput
- Streaming ingestion (Kafka/Redis) and multilingual sources
- Counterparty exposure (PFE) and expected-shortfall metrics

github.com/Ashish-1506/RiskPulse